

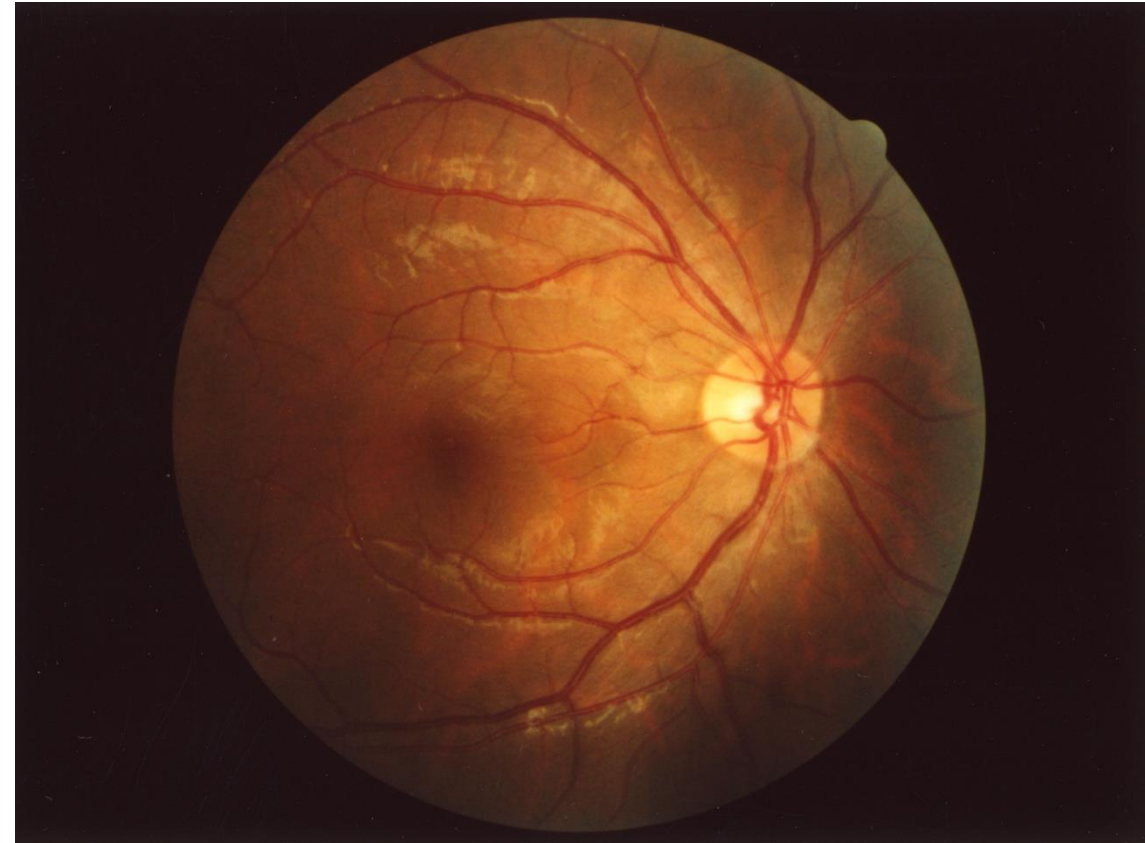
Development of a versatile auto-balanced Photodetector

22 April 2026

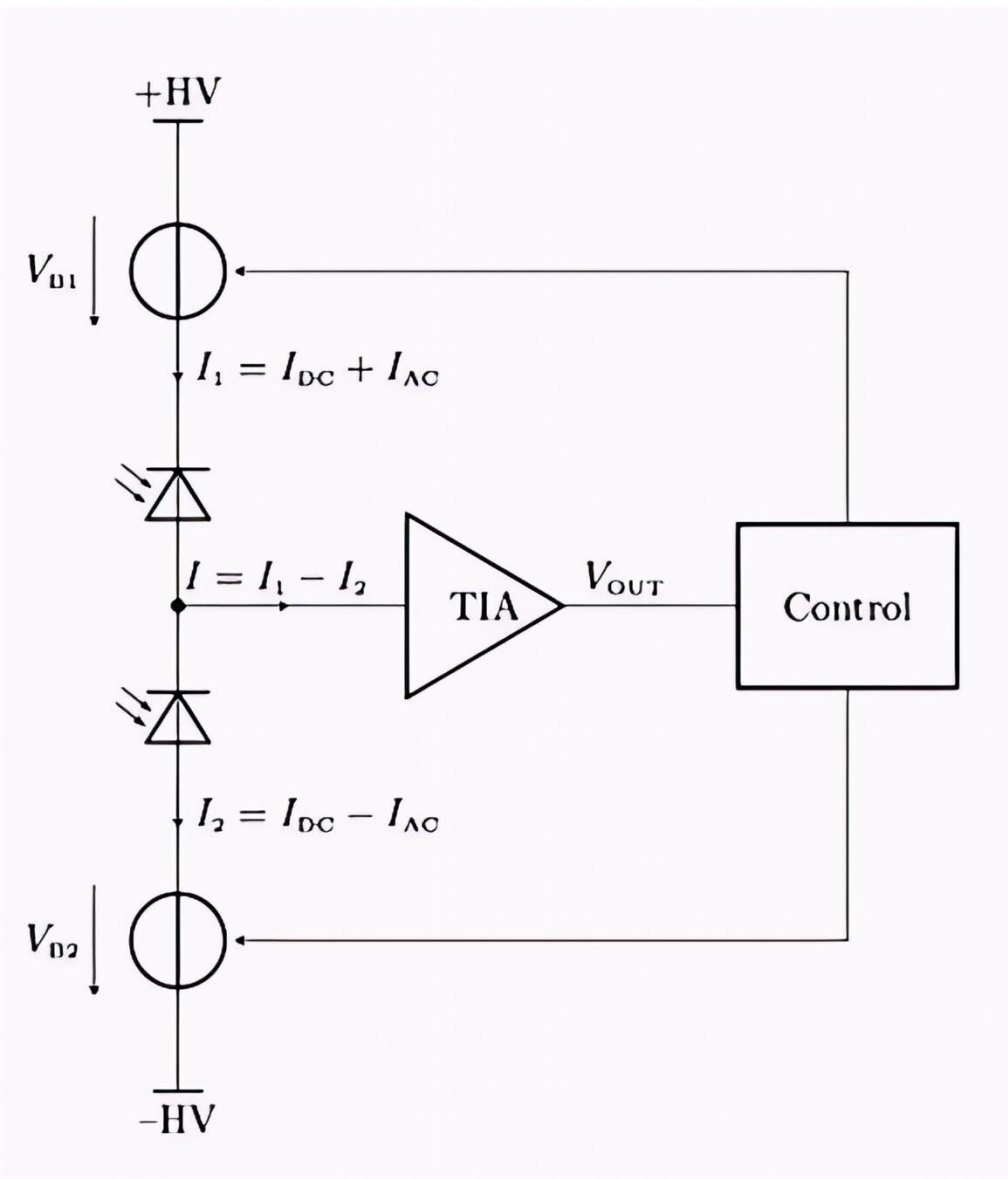
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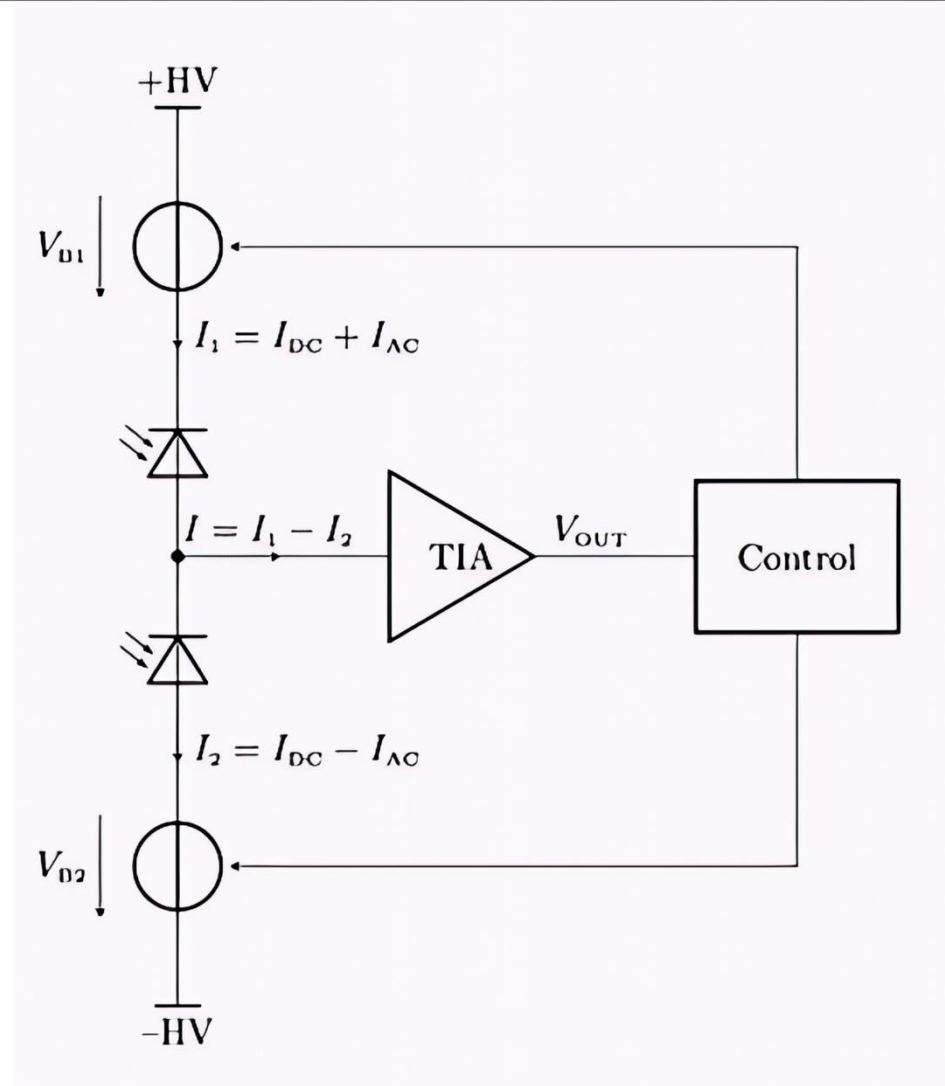
- **Laser noise cancellation spectroscopy**
weak absorption signal on a noisy laser
- **Optical coherence tomography (OCT)**
interferometric, strongly common-mode dominated
- **LIDAR front-ends** small reflected signal plus large background light
- **Quantum / low-light detection** single-photon-level signals, laser excess noise dominates



An ophthalmogram, an image of the fundus of an eye, of a healthy human male (Source Wikipedia [1])



- $I = I_1 - I_2 = 2 \cdot I_{AC} \rightarrow$ common-mode cancels
- I_1 and I_2 are **180° Phase shifted**
- **PIN (no gain) vs. APD** (internal gain M , needs stable reverse bias)
- **Wavelengths / materials** (Si, InGaAs, ...)
- **Bandwidths** from ~ 1 MHz up to 100MHz
- Wide range of effective responsivities
- One platform for all $\rightarrow$ topology must be **flexible**



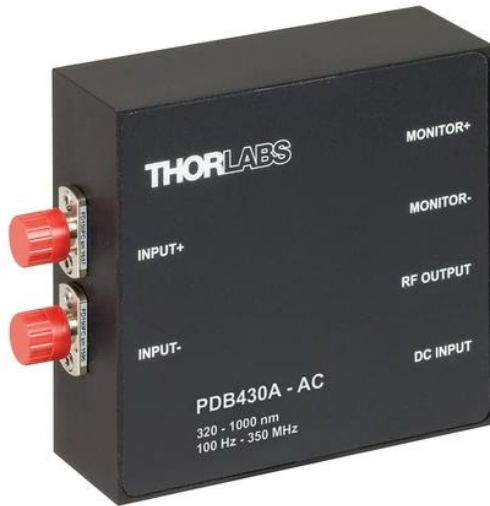
- **Two photodiodes stacked between +HV and -HV** — one signal, one
- **Currents:** $I_1 = I_{DC} + I_{AC}$ and $I_2 = I_{DC} - I_{AC}$ (antiphase AC components)
- **Summing node:** $I = I_1 - I_2 = 2 \cdot I_{AC} \rightarrow$ common-mode DC cancels; only signal enters TIA
- **Control block** $\rightarrow V_{B1}, V_{B2}$: senses the state of the front-end and drives the two bias sources this is the **bootstrap / auto-balance mechanism**

Active bias control is what makes the same topology usable across different diode types

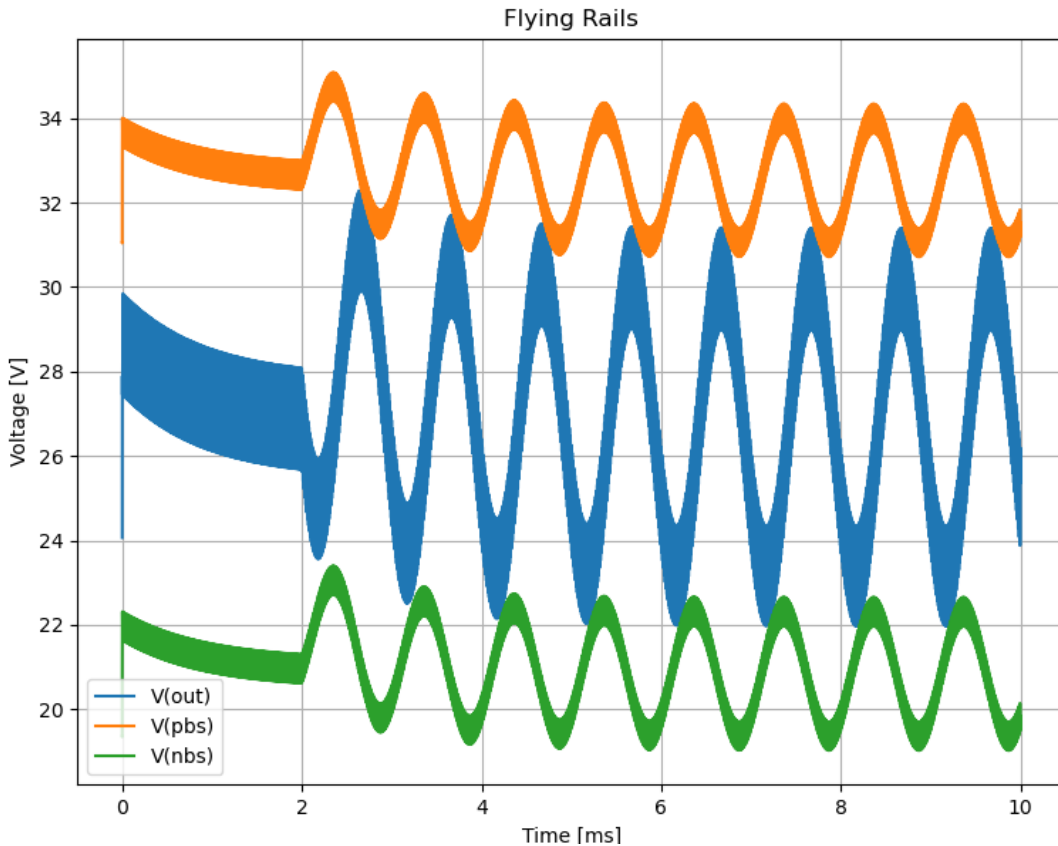
- Photocurrents span orders of magnitude across the supported diode types — large dynamic range
- Control loop must be **fast enough** not to attenuate the signal, **slow enough** not to destabilise the TIA
- Diode capacitance and parasitics directly affect loop stability and bandwidth
- Achievable CMRR depends on **matching** *and* on **loop bandwidth**
- Wide swing stresses op-amp supply range and dynamic behaviour

Commercial grade detectors

Device	CMRR (typ.)	Bandwidth (-3 dB)	Detector Type	Source
Thorlabs PDB430A / PDB430C	>20 dB (typ. >25 dB)	DC – 350 MHz	Si / InGaAs PIN	Thorlabs PDB4xx Series Datasheet[2]
Koheron PD100B (auto-balanced)	>30 dB	up to ~200 MHz	InGaAs	Koheron, PD100B Datasheet [3]



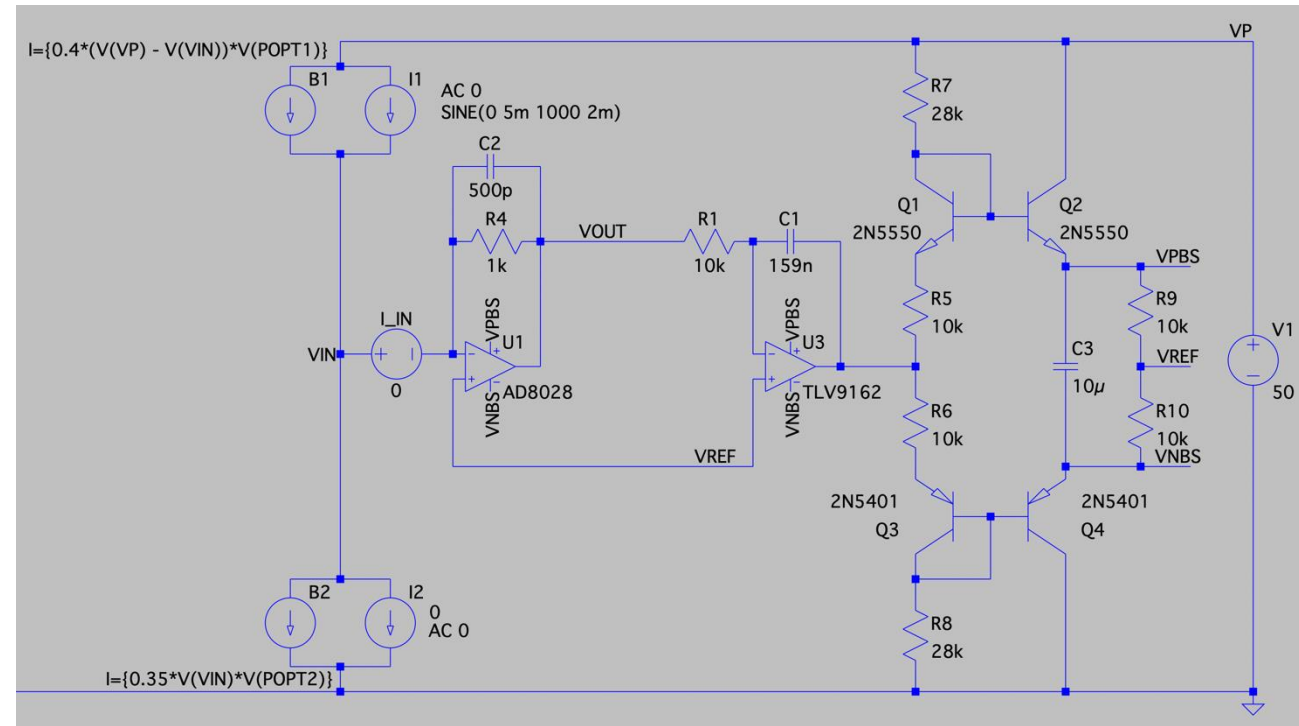
Balanced photodetectors used for differential signal detection. Left: Thorlabs [4], Right: Koheron PD100B[5]

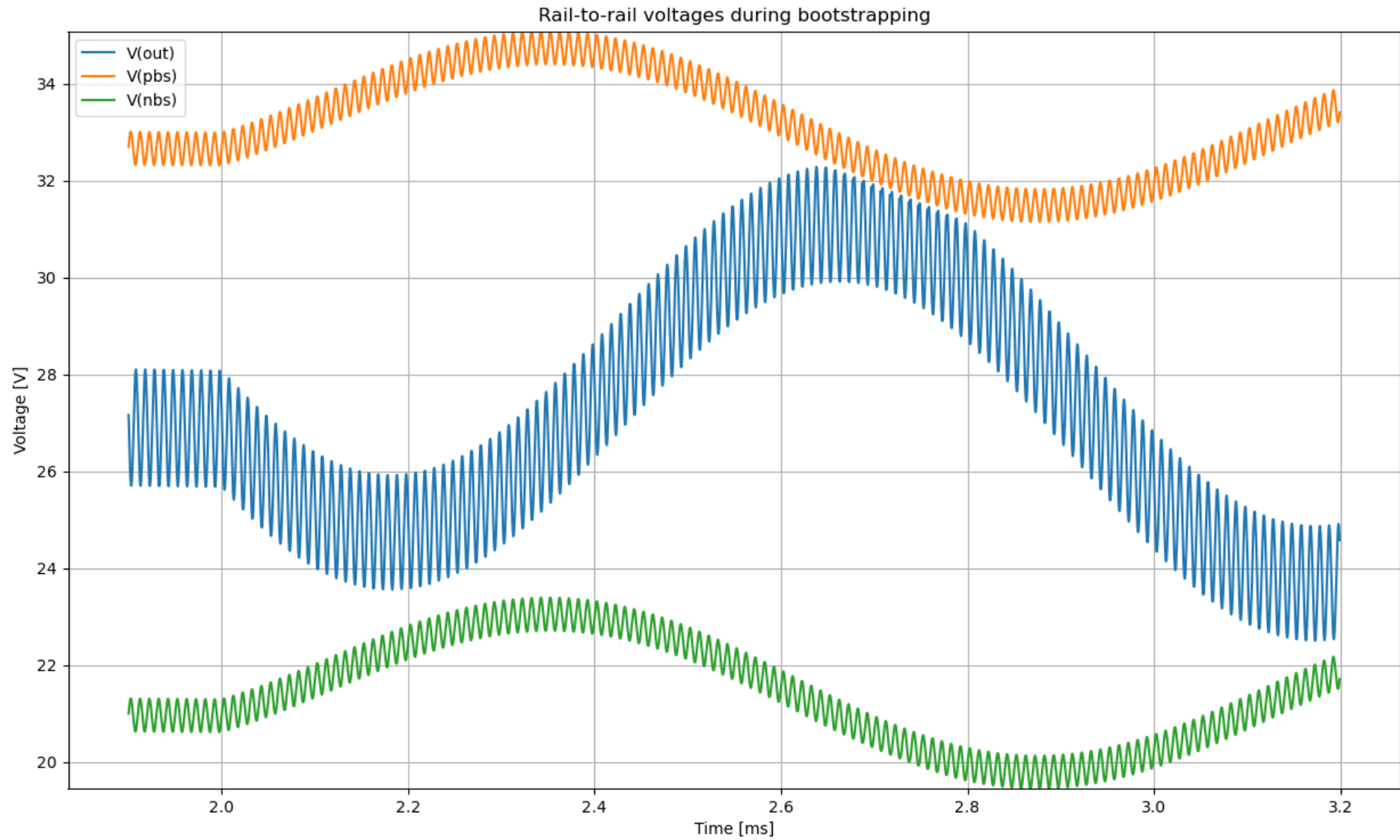


- **Flying rails principle:** local op-amp supply follows V_{out} ; large effective swing from low-voltage op-amp (*AN-1593 [6]*, *Jung EDN [7]*)
- **Loop speed > signal speed:** FET gate C + bias R form a low-pass; decoupling goes between transistor bases, not to ground
- **Op-amp criteria:** supply range, CM close to rails, slew rate, low bias current, no phase reversal (*TLV9162: 16 V max*)
- **High-impedance care:** guarding, shield bootstrapping, dielectric choice (*AN-241 [8]*)
- **Proven in hardware:** 50 Vpp @ 1 MHz with 3 op-amps; series R for capacitive loads (*TIDUEW8 [10]*)

- **Main question:** Can a bootstrapbased auto-balanced front-end serve as a *single, flexible* topology for a wide range of photodiodes (PIN and APD) with useful CMRR and bandwidth?
 - How does the bootstrap loop bandwidth affect the achievable CMRR?
 - Does the APD reverse bias remain constant under bootstrapping?
 - How do diode capacitance and parasitics influence loop stability?
 - Can the auto-balance mechanism compensate asymmetries without distorting the signal?
- **Scope / working assumptions:**
 - Target different PIN / APD diodes, modelled as current source with || capacitance
 - Bandwidth target: 1–10 MHz initially; extension to hundreds of MHz out of scope
 - Focus: topology & dynamic behaviour, not a specific optical application

- **Setup:** TIA (AD8028) $R_f = 1\text{ k}\Omega$, $C_f = 500\text{ pF} \rightarrow$ BW $\approx 318\text{ kHz}$; autobalance integrator (TLV9162) $\tau \approx 1.59\text{ ms} \rightarrow$ corner $\approx 100\text{ Hz}$
- **Diode models:** current sources with deliberate asymmetry (0.4 vs. 0.35 A/W) to test balancing
- **Observation 1:** autobalance loop nulls the DC offset between the two diode currents as expected
- **Observation 2:** 100 kHz probe signal propagates across internal nodes (V_PBS, V_NBS) a consequence of loop dynamics
- **Still open:** TIA gain sizing for realistic APD currents; quantitative frequency-response analysis of bootstrap loop





■ Design

- Dimension TIA gain for realistic APD photocurrents
- Frequency-response design of bootstrap / autobalance loop analyse two nested loops
- Quantify CMRR as a function of loop bandwidth
- Evaluate op-amp candidates against supply / slew / phase-reversal criteria

■ Build

- Perfboard → PCB implementation with selected op-amps and one PIN + one APD from available devices

■ Measure

- Characterise CMRR, bandwidth, bias stability under optical excitation
- Cross-validate against simulation predictions

- [1] Wikimedia Commons, 'Fundus of eye (normal)', https://commons.wikimedia.org/wiki/Category:Ocular_fundus#/media/File:Fundus_of_eye_normal.jpg
- [2] Thorlabs Inc., PDB4xx Series Datasheet, <https://www.manualslib.com/manual/2916657/Thorlabs-Pdb4-Series.html>
- [3] Koheron, PD100B Photodetector, <https://www.koheron.com/photonics/pd100b/>
- [4] Thorlabs Inc., "PDB430A-AC Balanced Photodetector", <https://www.thorlabs.com/item/PDB430A-AC>, accessed 21. April 2026.
- [5] Koheron, "PD100B Photodetector", <https://www.koheron.com/photonics/pd100b-photodetection/>, accessed 21. April 2026.
- [6] W. Jung, "Bootstrapping your op amp yields wide voltage swings", EDN Network, 2006.
- [7] Texas Instruments, "AN-1593: Extending Op Amp Operating Range via Bootstrapping" (SNAA049).
- [8] Texas Instruments, "AN-241: Working with High Impedance Op Amps" (SNOA664).
- [9] Texas Instruments, "AN-20: An Applications Guide for Op Amps" (SNOA621).
- [10] Texas Instruments, "Bootstrap High-Output Voltage-Extension Reference Design" (TIDUEW8), 2020.
<https://www.ti.com/lit/ug/tiduew8/tiduew8.pdf>